PM Surya Ghar Data Science Project

Step-by-step guideline for student teams

Purpose: Treat PM Surya Ghar as a pure data science and policy analytics project. The team should build a clean state-level panel dataset, perform forecasting, design scenario simulations, and produce a policy-facing dashboard or paper.

Recommended team structure: data engineering, exploratory analysis, forecasting/modeling, policy simulation, dashboard/reporting.

Preferred unit of analysis: State/UT x month or State/UT x quarter.

Minimum output: cleaned dataset, reproducible notebook, dashboard, and short research-style report.

1. Project framing

Core problem statement: What explains variation in PM Surya Ghar adoption across Indian states and districts, and how can future uptake be forecast under alternative policy scenarios?

Suggested student research questions:

- How fast is rooftop solar adoption growing across states, and which states are ahead or behind after adjusting for population, tariffs, and solar potential?
- Can installations, beneficiary counts, or subsidy disbursement be forecast reliably for the next 6 to 12 months?
- Which external factors are most strongly associated with uptake: tariff levels, solar resource, grid conditions, state income, or existing renewable energy base?
- What happens under counterfactual scenarios such as a 10% subsidy increase, a rise in electricity tariffs, or stronger rooftop suitability?

2. What data should be used first

Start with official PM Surya Ghar and rooftop solar sources. These should be treated as the backbone dataset.

Dataset / portal	Why it matters	Suggested use	Freshness note
data.gov.in Surya / PMSG keyword pages	Aggregates official PM Surya Ghar datasets in one place.	Discovery and download hub	Keyword pages show recently published resources
State-wise registrations, applications, households benefited	Useful for funnel analysis from interest to actual benefit.	Panel construction	Published Jul 2025 on OGD
State-wise installations / progress resources	Direct outcome variable for forecasting and growth measurement.	Target variable	Multiple 2025 resources on OGD/AP OGD
District-wise beneficiary datasets	Useful for district pilots and Maharashtra-focused analysis.	Geo drill-down	State-specific OGD resources
National Portal for Rooftop Solar	Official operational portal with DISCOM information and releases.	Context + cross-checks	Operational portal, ongoing

3. External datasets to combine with PM Surya Ghar

The strongest student projects will not stop at the scheme data alone. They will build a merged explanatory dataset with energy, tariff, solar resource, and macroeconomic signals.

External source	Variables to use	Why combine it	Recommended	Priority

			merge grain	
NITI ICeD solar potential	state solar potential, installed vs potential	normalizes adoption by resource endowment	state-year	High
MNRE physical progress	state RE installed capacity, rooftop context	captures renewable ecosystem maturity	state-month or state-year	High
CEA dashboard / power supply	demand, peak, supply position	grid stress and power demand proxies	state-month	High
Electricity tariff dataset / tariff book	domestic tariff, average rate	economics of rooftop payback	state-year	High
NASA POWER / Global Solar Atlas	irradiance, temperature, PV potential	resource suitability and seasonality	district-month or state-month	High
MOSPI / NITI economic deep dive	GSDP, income proxies, urbanization-related indicators	affordability and diffusion capacity	state-year	Medium

4. Recommended step-by-step workflow

1. **Step 1 - Data inventory:** Download all PM Surya Ghar resources, note publication date, update date, grain, and variable definitions. Create a dataset register first; do not start modeling before this.
2. **Step 2 - Build a canonical panel:** Convert all scheme datasets into a single long table with columns such as state, month_or_quarter, registrations, applications, installations, households_benefited, subsidy_amount, source_file, and update_date.
3. **Step 3 - Add external covariates:** Merge tariff, solar potential, state renewable capacity, electricity demand, and macroeconomic variables. Keep a data dictionary and explicit unit conversions.
4. **Step 4 - Clean and normalize:** Create per-lakh-population and per-1000-household measures. Compute lag variables, month-on-month growth, rolling averages, and subsidy per household.
5. **Step 5 - Exploratory analysis:** Create trend charts, state rank tables, funnel conversion plots, seasonality checks, and choropleths. Identify missingness and abrupt breaks caused by reporting updates.
6. **Step 6 - Forecasting:** Start with transparent baselines before advanced models. Benchmark naive, seasonal naive, Holt-Winters, ARIMA/SARIMA, Prophet, and gradient-boosted regression with lags and exogenous features.
7. **Step 7 - Policy analysis:** Run scenario simulations for subsidy changes, tariff changes, and potential-led targeting. Clearly separate predictive scenarios from causal claims.

8. **Step 8 - Communicate findings:** Prepare a dashboard and a short report with policy implications, uncertainty bands, limitations, and state-specific insights.

5. Forecasting tasks students can attempt

- Univariate forecasting: forecast state-wise installations, households benefited, or subsidy released using only historical scheme data.
- Multivariate forecasting: include tariff levels, irradiance, demand, and renewable capacity as exogenous regressors.
- Classification: predict whether a state will move into the top quartile of adoption in the next period.
- Panel forecasting: pooled model across all states with state fixed effects or grouped machine-learning models.
- Nowcasting: use near-real-time solar and weather variables plus recent portal updates to estimate current-quarter progress before official publication.

Minimum model comparison set:

- Naive and seasonal naive baselines
- Holt-Winters exponential smoothing
- ARIMA or SARIMAX
- Prophet only as a benchmark, not as the sole model
- XGBoost / LightGBM with lags, rolling means, and exogenous variables
- Optional: panel regression or Bayesian hierarchical time-series if the team is advanced

6. Policy-driven analyses that are worth doing

These should be framed as scenario analysis unless students use a defensible causal design.

Analysis idea	How to execute
Subsidy increase by 10%	Estimate elasticity from historical variation across states or time; simulate change in installations, households benefited, and subsidy burden.
Targeting high-potential but low-adoption states	Identify states with high solar potential but weak current uptake; propose differentiated incentive strategies.
Tariff shock analysis	Measure whether higher residential tariffs are associated with faster adoption because rooftop economics become more attractive.
DISCOM readiness segmentation	Cluster states or DISCOM contexts by demand, renewable penetration, and adoption funnel efficiency.
Adoption gap analysis	Compare actual adoption to expected adoption given solar potential, tariffs, and income proxies.
Rooftop suitability index	Build a composite index combining solar potential, urbanization proxy, tariff levels, and renewable ecosystem maturity.

7. Example scenario design

- Scenario A - Base case: forecast using observed subsidy and tariff levels.
- Scenario B - Subsidy +10%: adjust subsidy-related feature and simulate expected lift in installations or households benefited.
- Scenario C - High solar potential targeting: prioritize states in the upper band of solar resource but below-median adoption.
- Scenario D - Tariff increase: test how payback-sensitive states respond under higher residential tariffs.
- Scenario E - Combined intervention: subsidy +10% plus targeting of high-potential states.

  Important caution: a scenario model is not automatically causal. Students should say 'predicted under model assumptions' unless they use methods such as difference-in-differences, instrumental variables, synthetic control, or quasi-experimental identification.

8. Feature engineering suggestions

- Lagged installations: t-1, t-2, t-3
- Lagged subsidy amount and subsidy per beneficiary
- Application-to-installation conversion rate
- Registrations-to-application conversion rate
- Installations per lakh population
- Installations per 1000 electrified households
- Solar potential utilization ratio = actual rooftop adoption / estimated state potential
- Tariff-adjusted payback proxy = function of tariff, subsidy, and irradiance
- Seasonality features: month, quarter, summer indicator, monsoon indicator
- Renewable ecosystem maturity proxy = state RE capacity share or rooftop share

9. Validation and reporting standards

- Use walk-forward validation or rolling-origin evaluation instead of random train-test splits for forecasting.
- Report MAE, RMSE, MAPE or sMAPE, and compare every model against a naive baseline.
- Provide calibration plots or prediction intervals wherever possible.
- Show one national-level result and at least five representative state case studies.
- Document data freshness clearly because PM Surya Ghar resources are updated in batches rather than through a clean public real-time API.

10. Deliverables for student teams

- Dataset register and data dictionary
- Merged master CSV or parquet file
- Reproducible Python notebook or script pipeline
- Forecast comparison table
- Policy scenario analysis table
- Interactive dashboard in Tableau, Power BI, or Streamlit

- 4 to 6 page report written in research style

11. Where this work can be published

Relevant conferences:

- ACM e-Energy - strong fit for data-driven forecasting, policy analytics, and future energy systems.
- IEEE Power & Energy Society General Meeting - good fit for power systems, rooftop solar, grid-facing analytics, and utility relevance.
- IEEE Photovoltaic Specialists Conference (PVSC) - good fit if the paper leans more heavily into photovoltaic deployment, techno-economics, and solar resource analysis.

Relevant journals:

- Applied Energy - best if the work is methodologically strong and energy-system focused.
- Energy Policy - best if the paper is framed around subsidy design, adoption barriers, and policy scenarios.
- Solar Energy - strong fit for solar deployment, techno-economics, and solar potential analysis.
- Energy for Sustainable Development - good fit for India-focused sustainable energy access and policy work.

12. Useful links: datasets, portals, reports, and venues

PM Surya Ghar / OGD dataset hubs

https://www.data.gov.in/keywords/Surya

https://www.data.gov.in/keywords/PMSG

https://www.data.gov.in/resource/state-ut-wise-number-registrations-applications-and-households-benefitted-under-pm-surya

https://ap.data.gov.in/resource/stateut-wise-number-installations-under-pm-surya-ghar-muft-bijli-yojana-9-12-2024

https://www.data.gov.in/resource/district-wise-number-beneficiaries-under-pm-surya-ghar-muft-bijli-yojana-state-maharashtra

https://solarrooftop.pmsuryaghar.gov.in/

Energy, tariff, and potential data

https://iced.niti.gov.in/energy/fuel-sources/solar/potential

https://iced.niti.gov.in/analytics/state-wise-deep-dive

https://cea.nic.in/dashboard/?lang=en

https://cea.nic.in/power-supply/?lang=en

https://mnre.gov.in/en/physical-progress/

https://www.data.gov.in/resource/state-wise-average-rate-electricity-domestic-and-industrial-consumers

https://www.data.gov.in/resource/state-wise-potential-various-renewable-energy-technologies

https://www.mospi.gov.in/sites/default/files/publication_reports/Energy_Statistics_2025/Energy%20Statistics%20India%202025_27032025.pdf

Solar resource and weather

https://power.larc.nasa.gov/data-access-viewer/

https://power.larc.nasa.gov/docs/services/api/temporal/monthly/

https://globalsolaratlas.info/

https://globalsolaratlas.info/downloads

Reports and background reading

https://ieefa.org/resources/advancing-residential-rooftop-solar-adoption-india-under-pm-surya-ghar-yojana

https://www.pib.gov.in/PressReleasePage.aspx?PRID=2199729&lang=1&reg=3

https://link.springer.com/article/10.1186/s42162-025-00562-0

https://academic.oup.com/ce/article/7/5/994/7273023

Publication venues

https://energy.acm.org/conferences/eenergy/2026/

https://pes-gm.org/

https://ieee-pvsc.org/

https://www.sciencedirect.com/journal/applied-energy

https://shop.elsevier.com/journals/subjects/physical-sciences-and-engineering/energy-and-power?page=3&subjectArea=physical-sciences-and-engineering%2Fenergy-and-power

https://www.ises.org/what-we-do/publications/solar-energy-journal

https://www.sciencedirect.com/journal/energy-for-sustainable-development

13. Recommended student project timeline

Week	Focus	Expected output
1	Data inventory and download	Dataset register, raw files, source notes
2	Cleaning and schema design	Canonical master table and data dictionary
3	EDA and visualization	State comparisons, trends, missingness report
4	Baseline forecasting	Naive, exponential smoothing, ARIMA benchmarks

5	Feature enrichment	Tariff, solar potential, demand, and macro merges
6	Advanced models and scenarios	XGBoost/SARIMAX plus subsidy and tariff scenarios
7	Dashboard and writing	Policy visuals, insights, reproducible notebook
8	Final review	Presentation, report, repository cleanup

14. Final advice to students

Do not treat this as only a machine-learning exercise. The strongest projects will combine careful data engineering, transparent forecasting baselines, policy-aware interpretation, and clear communication of uncertainty. The final output should be understandable to both a data science audience and an energy-policy audience.